

Multiobjective Tree-Structured Parzen Estimator

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Motivation: Real-World Problems

- ▶ Many real-world problems involve optimizing **multiple conflicting objectives** simultaneously

Examples:

- ▶ Mechanical design (diesel engine, air intake systems) requiring expensive CFD simulations
 - ▶ Automated ML: high accuracy **vs.** low computational cost
- ▶ Objective functions are:
 - ▶ **Expensive** to evaluate (time/money)
 - ▶ **Black-box** (no analytical form)
 - ▶ Defined on **complex/tree-structured** search spaces (mix of real, integer, categorical + conditional parameters)
- ▶ Need an algorithm that:
 - ▶ Works with a **limited evaluation budget**
 - ▶ Handles **complex search spaces** naturally
 - ▶ Is **scalable** and **parallelizable**

Limitations of Existing Methods

Gaussian Process (GP)-based methods (PESMO, SMS-EGO, ParEGO):

- ▶ **De facto standard** in Bayesian optimization
- ▶ Provide uncertainty estimates for exploration/exploitation trade-off
- ▶ **But:**
 - ▶ Not suitable for non-continuous/conditional spaces
 - ▶ High computational complexity $O(n^3)$ — poor scalability
 - ▶ Approximation techniques help but cause performance degradation

Tree-Structured Parzen Estimator (TPE):

- ▶ Non-GP surrogate (Parzen window density estimation)
- ▶ Naturally handles complex/conditional spaces
- ▶ Scales to ~ 1000 observations, tens of variables
- ▶ Outperforms GP-based methods on single-objective HPO
- ▶ **But:** Designed for **single-objective** only!

Formal Problem Statement

Problem:

$$\underset{x \in X}{\text{minimize}} \quad f(x) := (f_1(x), \dots, f_m(x))$$

Given:

- ▶ Tree-structured search space $X = X_1 \times \dots \times X_n$
- ▶ Mixed variable types: real, integer, categorical
- ▶ Conditional variables (active only when parent takes specific values)
- ▶ Expensive black-box objectives f_j

Goal: Efficiently approximate the **Pareto front** with limited evaluation budget

Dominance (minimization):

- ▶ $y \prec y'$ iff $\forall i : y_i \leq y'_i$ and $\exists i : y_i < y'_i$
- ▶ Pareto set: $\{x \in X \mid \nexists x' \in X : f(x') \prec f(x)\}$

Background: TPE for Single-Objective

Key Idea: Model $p(x_i | y)$ using two densities:

$$p(x_i | y) = \begin{cases} l(x_i) & \text{if } y < y^* \quad (\text{"good" observations}) \\ g(x_i) & \text{if } y \geq y^* \quad (\text{"poor" observations}) \end{cases}$$

where y^* is quantile threshold ($p(y < y^*) = \gamma$)

Acquisition Function (Expected Improvement):

$$\text{EI}_{y^*}(x_i) \propto \left(\gamma + (1 - \gamma) \frac{g(x_i)}{l(x_i)} \right)^{-1}$$

$\Rightarrow$ Maximize EI by maximizing ratio $l(x_i)/g(x_i)$

Density Estimation (per parameter):

- ▶ Real/Integer: truncated Gaussian (adaptive bandwidth)
- ▶ Categorical: weighted histogram
- ▶ Conditional variables sampled from root to leaves

MOTPE: Key Contributions

Two Main Challenges in Extending TPE:

1. **Splitting observations** with multiple incomparable objectives
2. **Multiobjective acquisition function** that drives Pareto front improvement

Solution Overview:

- ▶ **Split rule:** $y \prec Y^*$ or $y \parallel Y^*$ for $l(x_i)$, $Y^* \preceq y$ for $g(x_i)$
- ▶ **Greedy splitting:** Nondominated sorting + Hypervolume Subset Selection (HSSP)
- ▶ **Weighting:** Assign weights proportional to hypervolume contribution for $l(x_i)$
- ▶ **Acquisition:** Expected Hypervolume Improvement (EHVI)
 $\propto \left(\gamma + (1 - \gamma) \frac{g}{l} \right)^{-1}$

MOTPE: Splitting Observations (Algorithm 2)

Goal: Select $\lfloor \gamma |D| \rfloor$ observations for $l(x_i)$ (good set)

Greedy Algorithm:

1. Compute nondomination ranks (Pareto sorting)
2. Greedily add full nondomination fronts while $|D_l| + |D_{\text{rank}(j)}| \leq \lfloor \gamma |D| \rfloor$
3. For remaining slots, solve **Hypervolume Subset Selection Problem (HSSP)**

HSSP via Greedy (Algorithm 3):

- ▶ Find subset of size n_s maximizing hypervolume indicator
- ▶ Greedy algorithm gives $(1 - 1/e)$ -optimality guarantee (submodular maximization)
- ▶ Reference point: $(\max(y_1, y'_1), \dots, \max(y_m, y'_m))$

Weighting: $w_j \propto$ hypervolume contribution for $l(x_i)$, uniform weight 1 for $g(x_i)$

MOTPE: Algorithm Overview (Algorithm 4)

Input: Observations D , quantile γ , candidates n_c

For each iteration:

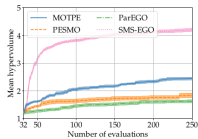
1. Split D into D_l and D_g (Algorithm 2)
2. **For each active parameter x_i (root to leaves):**
 - ▶ Construct densities $l(x_i)$ from D_l , $g(x_i)$ from D_g
 - ▶ Sample n_c candidates from $l(x_i)$
 - ▶ Select $x_i^* = \operatorname{argmax} l(x_i)/g(x_i)$
3. Evaluate $f(x^*)$, add to D

Complexity:

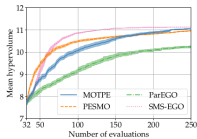
- ▶ Memory: $O(kn)$ where k = observations, n = parameters
- ▶ Time per iteration: $O(k \log k + n_c \cdot n \cdot \text{cost}(l))$
- ▶ Much faster than GP-based methods (hours vs. days in experiments!)

Experiment 1: WFG Benchmark (Low Dim, Limited Budget)

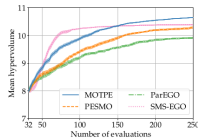
Setup: Comparison with GP-based methods (PESMO, ParEGO, SMS-EGO) on WFG problems.



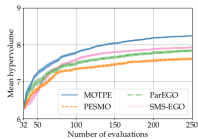
(a) WFG1



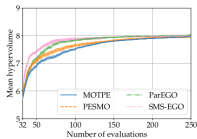
(b) WFG2



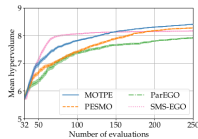
(c) WFG3



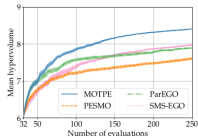
(d) WFG4



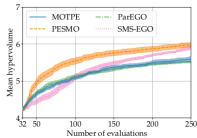
(e) WFG5



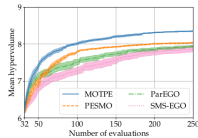
(f) WFG6



(g) WFG7



(h) WFG8



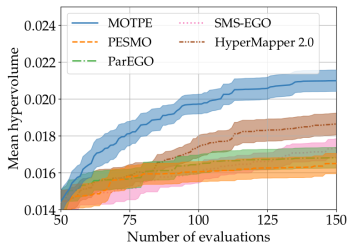
(i) WFG9

Experiment 2: Real-World Problem — CNN Design

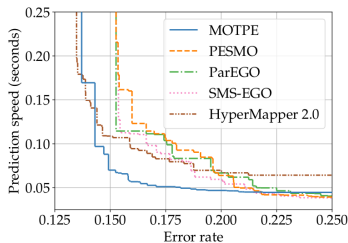
Task: Design CNN for CIFAR-10 with two objectives:

1. Minimize classification **error rate**
2. Minimize **prediction time** (inference speed)

Search space: 13 parameters (many conditional), including number of blocks, filters, batch norm, etc.



(a) Mean hypervolume $\pm$ standard error.



(b) 50%-attainment surface.